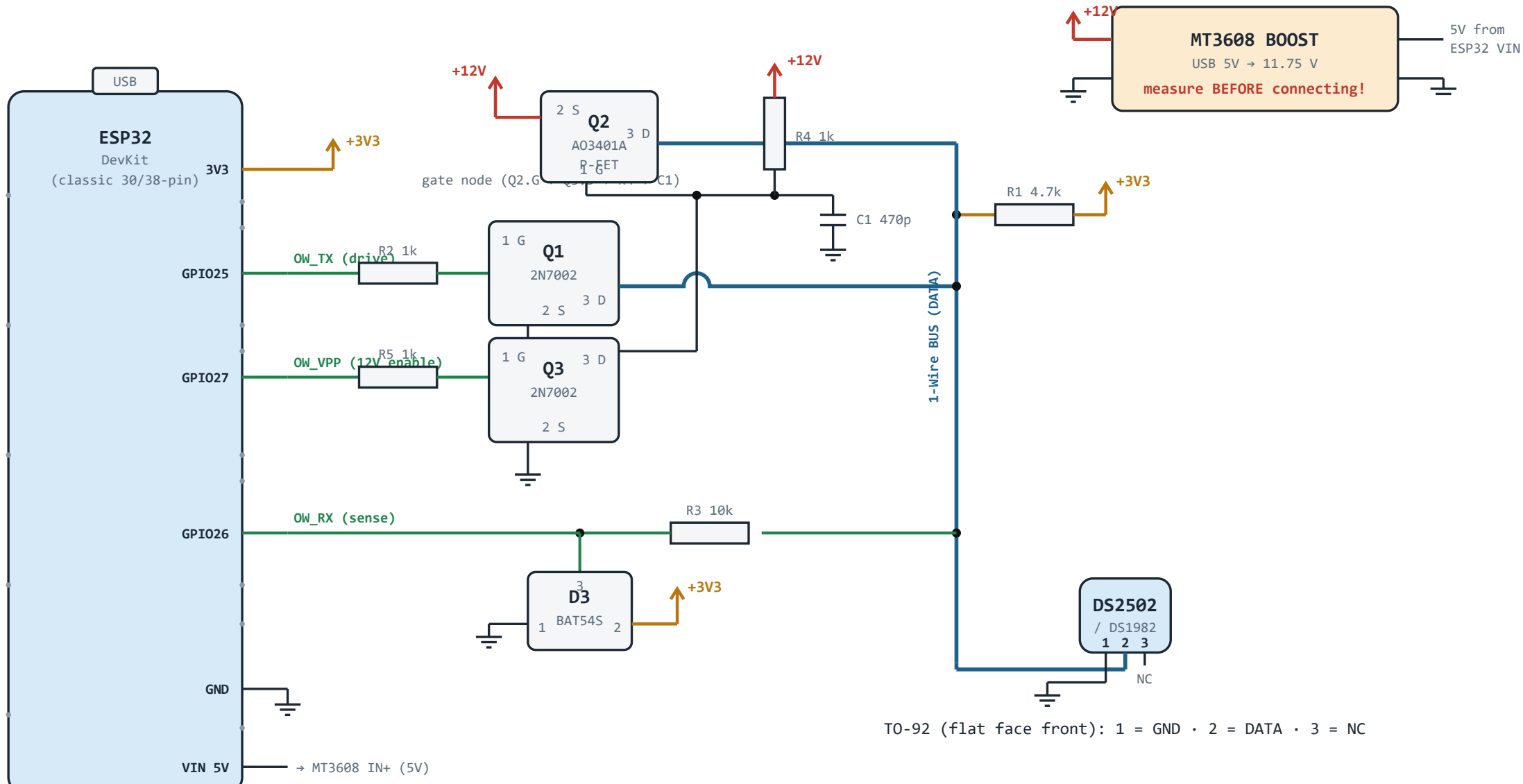


DS2502 Clone Tool – ESP32 Bridge: Pin Connection Diagram

driver circuit build · firmware defaults OW_TX=GPI025 · OW_RX=GPI026 · OW_VPP=GPI027 · circuit theory: docs/schematic.svg



DS2502 other packages – SO-8: DATA = pin 3, GND = pin 4 · TSOC: GND = 1, DATA = 2 · SOT-23: DATA = 1, GND = 2 & 3 · WLP: 1A/1B = DATA, 2A/2B = GND
 SOT-23 top views (pin 1 left): 2N7002 → 1=G, 2=S, 3=D · A03401A → 1=G, 2=S, 3=D · BAT54S → 1=A1, 2=K2, 3=K1·A2 (series junction)
 BAT54S wiring: pin 3 → GPI026-side node of R3 · pin 1 → GND · pin 2 → 3V3. ESP32-S2/S3/C3: see docs/PINOUT.md for pin choices (GPIO 22–25 absent on S2/S3).
 ⚠ Set the MT3608 to 11.75 V (loaded) BEFORE connecting the DS2502 – VPP must be 11.5–12.0 V; 12.0 V is the absolute maximum.

— 1-Wire bus — GPIO / sense — +12 V (VPP) — +3.3 V — GND

Read-only quick build: OW_USE_DRIVER 0 → GPIO4 → bus + 4.7 kΩ pull-up to 3V3, GND common (no 12 V, no driver parts).